Lecture 2
The Goods Market

Economics B — Macroeconomics

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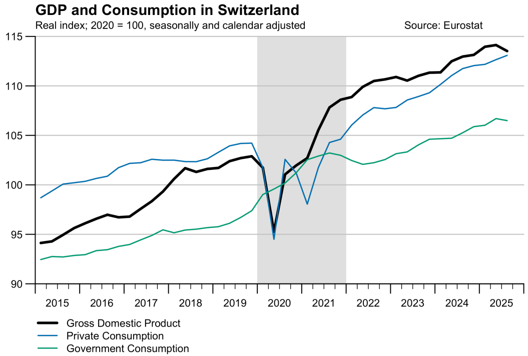
Motivation: The Corona Crisis

- Using the years 2020–2023, marked by the Corona crisis, we can illustrate many macroeconomic issues with real-world examples.
- Private consumption and GDP fell — at the same time, government spending increased.



Contents of this Lecture

How Economists Analyze Economic Policy Problems

Using (mathematical) models:

- Precision of the problem, transparent assumptions, consistent analysis.
- Clear indication of model assumptions and causal relationships.
- Basic concept of an economic model:
  - Mathematical functions describe causal relationships
  - Model assumptions influence both causality and dynamics
  - Exogenous (independent) and endogenous (dependent) variables
  - Model parameters
- The model shows dynamics and is suitable for comparative statics.
- Comparison of model statements with empirical analysis.

Empirical Macroeconomics

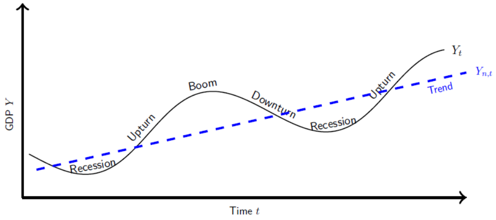
- Empirical analyses require experiences and real events.
- In macroeconomics there are no artificial laboratory data to test hypotheses. One must make do with the available data based on past real situations.
  - Example: Did the Hartz reforms reduce German unemployment?
- Important econometric insights are based on
  - Time series data
  - "Natural experiments" (e.g. German reunification)
  - Comparisons between countries (or states, cantons)

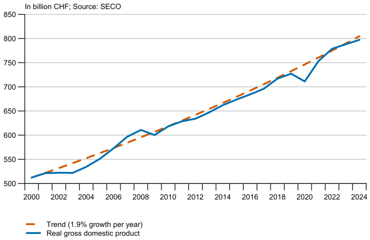
Learning Objectives and Literature of the 2nd Lecture

- Learning Objectives:
  - The economy in the short-run
  - Idea and construction of a macroeconomic model
  - Components of gross domestic product (GDP) and their determinants
  - Consumption theories according to Keynes and Friedman
  - Derivation of the IS curve for the IS-LM model
- Literature: Textbook by Blanchard: Chapter 3, 5.1, 14.1 & 14.2

GDP over Time & Course Structure

- In the long run GDP follows a (hopefully rising) path.
- At any given time there is a GDP value on the trend line ($Y_{n,t}$).





- We first examine short-term fluctuations ($\longrightarrow$ Lectures 2 to 4).
- Then, we determine $Y_{n,t}$ at a given time t ($\longrightarrow$ Lectures 5 to 7).
- Finally, we analyze the trend of $Y_{n,t}$ ($\longrightarrow$ Lectures 10 to 11).

The Economy in the Short-Run

- In lectures 2, 3 and 4 we focus on the *short*-run.
  - Short-run (< 5 years), medium-run (5-10 years), long-run (>10 years).
  - The subdivision is based on how quickly economic processes unfold.
  - Each requires a different perspective and economic analysis tools.
- Economic variables behave differently in the short-run than in the long-run.
  - Example: Government spending is increased. Short-term GDP increase; the long-term effect is much less clear.
  - In the short-run, we ignore resource constraints and analyze economic fluctuations. Prices and wages are rigid in the short-run. In the medium-run, price increases can lead to inflation, for example.

Schedule Lectures 2–4

- We want to analyze the economy within the framework of an economic (theoretical) model.
- Economic models consist of mathematical equations:
  - Advantage: assumptions and causal relationships clearly described, unlike the verbal 'anything goes' approach.
  - "Too many economists fall in love with the math and forget its instrumental nature." — Dani Rodrik (Economics Rules, p.35)
- Abstract and simplify economic models.
  - "A model that takes into account the whole colorfulness of reality, wouldn't be any more useful than a one-to-one map." — Joan Robinson

Schedule Lectures 2–4 (continued)

- We use the well-known IS-LM model:
  - As simple as possible, but not too simple.
  - We essentially need: goods market, financial market
    - We leave out for now: labor market, stocks, foreign countries, ...
  - Goods market in equilibrium (IS curve): supply and demand
  - Financial market in equilibrium (LM curve): supply and demand
- We describe supply and demand with simple mathematical formulas.
- Goal: Understanding model, applying model, questioning model.

An Example

- Japan raised the value added tax from 8% to 10% as of Oct 1, 2019.
  - On October 8th, the *Financial Times* wrote:
    "Japan data point to bigger than expected hit from sales tax rise
    Drop in spending adds to pressure on central bank to ease monetary policy"
  - How can the effects of a tax increase be explained?
  - Japan's government increased VAT knowing the short-term negative effects.
    The goal was to reduce the public deficit.
    How is the measure to be assessed in the medium-term?
- In analogy to other sciences, macroeconomics attempts to depict causal causal relationships using (mathematical) models.

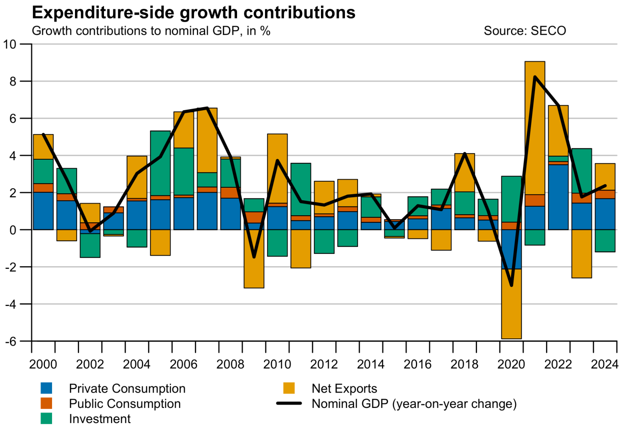
The Economy in the Short-Term

- With annual fluctuations in economic activity, the interrelation between production, income and demand is central to macroeconomic understanding:
  - Changes in demand lead to adjustments in production.
  - Adjustments in production trigger changes in income.
  - Changes in income cause changes in demand.
- In the analysis of the goods market we therefore focus on
  - Demand for goods, production, and income.
  - As seen in lecture 1, all three correspond to GDP.

Components of Gross Domestic Product (GDP)

- The GDP can be determined in three ways:
  - Production approach
  - Income approach
  - Expenditure approach
- GDP = Y = C + I + G + NX
  - C: Private consumption of goods and services
  - I: Private investment
  - G: Government purchases of goods and services
  - NX = X – IM: Net exports (exports minus imports)
- What factors determine C, I, G? (assumption: closed economy, i.e. $X = IM = 0$)

Swiss GDP Growth over Time



- Consumption fluctuates much less than GDP.

Theories on Consumption

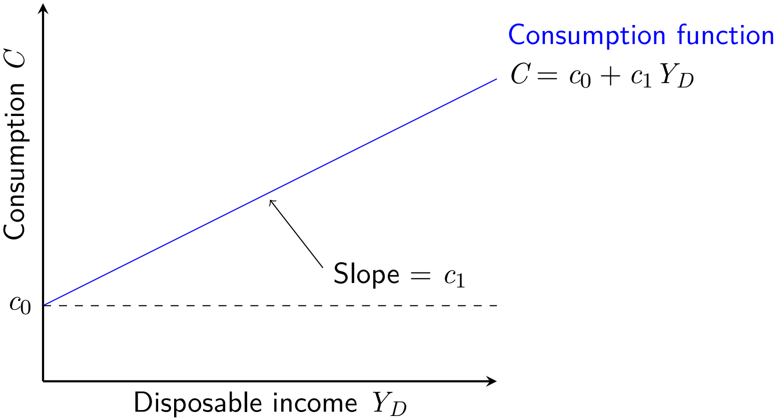
- How is the consumption C determined?
- Idea of the Keynesian consumption function:
  - Consumption is based only on current disposable income.
  - The propensity to save/consume is postulated from aggregated data (there is no micro-foundation).
- Idea of the permanent income hypothesis:
  - Households with a longer-term planning horizon choose consumption in such a way that that an intertemporal utility function is maximized (as a result we see consumption smoothing).
- We focus here on the Keynesian approach.
- $\longrightarrow$ Further discussion in the tutorials

Components of GDP — Consumption

- Consumption function: $C(Y_D)$ with Y_D as disposable income $Y_D = Y - T$:
  - $C = C(\underset{(+)}{Y_D})$ describes an economic context as a function,
    $C'(\underset{(+)}{Y_D}) > 0$ marginal propensity to consume
  - Note: $T =$ Taxes – transfer payments
- As linear function: $C = c_0 + c_1 Y_D = c_0 + c_1(Y - T)$
  - Two parameters:
    autonomous consumption c_0 and propensity to consume $0 < c_1 < 1$

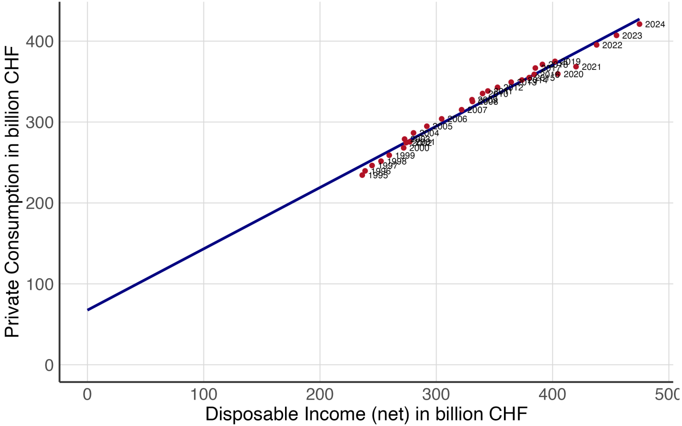
Components of GDP — Consumption (graphical)

- Graph:



Keynesian Consumption Function in the Data

- Based on Swiss data for income (Y_D) and consumption (C) the result is an estimated function $C = \hat{c}_0 + \hat{c}_1\, Y_D = 68 + 0.8\, Y_D$.



Source: Federal Statistical Office, je-d-04.02.01.02 and je-d-04.02.02.01

Keynesian Consumption Theory: Discussion

- Every economic model requires simplifying assumptions — it is important (after understanding of the model) to critically examine them.
- Fundamental critique of the Keynesian consumption function: it is postulated and not derived (not micro-founded).
- Important factors not considered:
  - Income shifts over the life cycle
  - Consideration of future income (and taxes)
  - Savings, interest rate
  - Wealth

Permanent Income Hypothesis

- The permanent income model describes how households, taking into account their budget constraints, determine their consumption path over time in order to maximize their utility.
- Elements of the model:
  - Lifetime income: all present and future income.
  - Utility function: Consumption and leisure generate utility.
  - Intertemporal budget constraint:
    Households cannot spend more than they earn.
  - Optimization within the constraints ($\longrightarrow$ *tutorials*)

Components of GDP — Investment

- Three components of private investment: investment in equipment by firms, investment in construction, changes in inventories
- Private investment is important for two reasons:
- (1) Part of GDP: Y = C + I + G + NX
  - Investments explain the level and (short-term) fluctuations in output.
  - Investment is much more volatile than consumption $\longrightarrow$ household consumption smoothing
- (2) Increase the stock of productive capital
  - $K_{\text{tomorrow}} = K_{\text{today}} + I$ – Depreciation
  - Investment central to long-term economic growth $\longrightarrow$ Lecture 10.

Components of GDP — Investment (continued)

- What determines the level of private investment?
- (1) The interest rate
  - Companies have many potential investment projects with different returns (*returns-on-investment*, *RoI*).
  - Every project has a net present value (*net present value*): $NPV = CF_0 + CF_1/(1+i) + CF_2/(1+i)^2 + CF_3/(1+i)^3 + \ldots$
    - CF_0 Investment today = (negative) cash flow
    - CF_1, CF_2, CF_3, ...: Future earnings (in case of uncertainty: expected earnings)
  - A lower interest rate makes more projects profitable ($\longrightarrow$ opportunity costs)
- (2) The level of production:
  - To be able to produce more, additional machines are necessary.

Components of GDP — Investment (continued)

- We summarize both aspects mathematically: $I = I(\underset{(+)}{Y}, \underset{(-)}{i})$
- In contrast to Blanchard, you often find $I(i)$.
- The model could be supplemented by other factors:
  - Expectations about the future ($\longrightarrow$ PMI)
  - Risk preferences
- In the textbook investments are sometimes fixed, i.e. they are assumed to be exogenous: $I = \bar{I}$. This simplifies the analysis.
  - Endogenous variables: depend on other variables in the model.
  - Exogenous variables: not explained in the model.

Components of GDP — The State

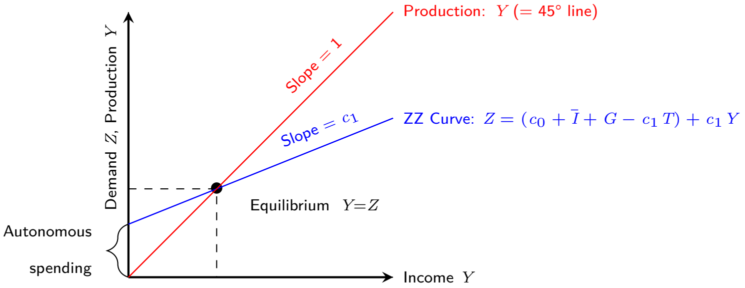
- GDP: Y = C + I + G + NX
- How can the state act influence GDP?
  - Direct and indirect taxes $\longrightarrow T$
  - Transfer payments (e.g., unemployment benefit) $\longrightarrow T$
  - Government spending on goods and services $\longrightarrow G$
- Decisions on government spending, G, and the amount of taxes and transfers, T, are called fiscal policy.
- We treat G and T as exogenous variables.

Equilibrium on the Goods Market

- An equilibrium on the goods market can only be achieved if
  if the production of goods, Y, equals the demand for goods, Z:
  - $Y = Z = c_0 + c_1 * (Y - T) + I(Y, i) + G$ (assumption: $X = IM = 0$)
  - In equilibrium, production Y (left side of the equation)
    equals demand (right side of the equation).
  - Demand in turn depends on income Y;
    income in turn is equal to production. Idea: Unused capacities.
  - Both production and income are denoted by Y.
  - In practice, Y and Z can differ from each other.

Equilibrium on the Goods Market — Graphical

- Equilibrium on the goods market with $I = \bar{I}$:
  - Supply of goods ($=$ production $=$ income $= Y$) $=$ demand for goods ($= Z$)
- Graph (Keynesian cross):



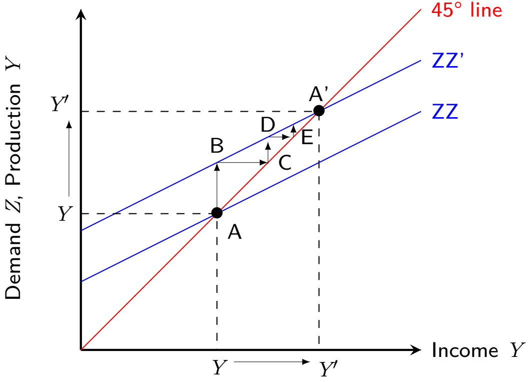
- Note: the ZZ curve is drawn for a given interest rate.

Equilibrium on the Goods Market — Comparative Statics

- The demand for goods $Z = (c_0 + I(Y,i) + G - c_1 T) + c_1 Y$ increases if
  - autonomous consumption (c_0) is increasing
  - the government spending (G) is increased
  - taxes (T) are reduced
  - the interest rate (i) is lower
- How does the equilibrium change if government spending G increases?
  - In equilibrium the following applies: $Y = Z = c_0 + c_1 * (Y - T) + I(Y,i) + G$
  - Let us assume $I = \bar{I}$ and solve the equation for Y:
    $Y = \frac{1}{1-c_1}\left[c_0 + \bar{I} + G - c_1 T\right] \quad \Longrightarrow \frac{1}{1-c_1}$ = multiplier
  - If G increases by 1, Y increases by $1/(1-c_1) > 1$

Equilibrium on the Goods Market — Comparative Statics

- An increase in government spending G



- Note: $\Delta Y = 1/(1 - c_1) * \Delta G > \Delta G$.

How Large is the Multiplier?

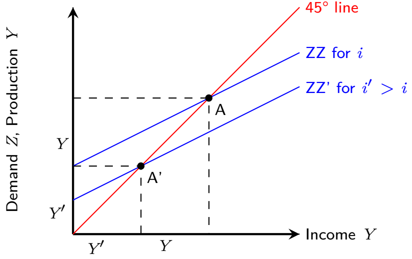
- $\Delta Y = 1/(1 - c_1) * \Delta G > \Delta G$
  - Valid only for unutilized capacities; hence the amount is often overestimated.
- Examples: HSG, EU Green Deal, Juncker Plan

Equilibrium on the goods market — Multiplier effect

- For a given increase in the ZZ curve, the rise of Y depends on how big c_1 is. Why?
  - If demand increases by 100 CHF, then after n rounds there is an increase in production of 100 CHF multiplied by the following sum: $1 + c_1 + c_1^2 + ... + c_1^n$. ($\longrightarrow$ geometric series)
- Numerical example: The state pays 100 CHF for road repair.
  - Road builder spends (at $c_1 = 0.8$) 80 CHF in a shop and saves 20 CHF.
  - Retailers spend 80*0.8=64 CHF and save 16 CHF.
  - After 2 rounds a cumulative increase of 100 + 80 + 64 = 244 CHF.
  - After all the rounds:
    Increase in GDP by 500 CHF (multiplier of $1/(1 - c_1) = 5$).

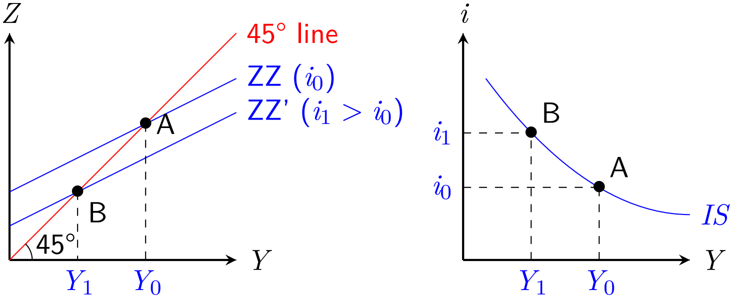
Derivation of IS curve

- The IS curve describes the equilibrium on the goods market.
- The effects of a rise in interest rates on income:
  - investments are interest-dependent, i.e. $I = I(i, Y)$.
  - A rise in the interest rate causes investments and thus income to decline. The demand curve for goods shifts downwards.



Derivation of IS curve (continued)

- As interest rates rise, income falls in the goods market equilibrium. The IS curve therefore has a downward slope:



- Above the IS curve: Excess supply on the goods market
  Below the IS curve: Excess demand on the goods market
  On the IS curve: equilibrium on the goods market

IS-Curve: Investments and Savings

- Name of the IS curve: *S̲avings* = I̲nvestments.
- Investment equals savings:
  An alternative approach to the equilibrium on the goods market.
- Private consumer savings: $S = Y - T - C$
- GDP equation (excluding exports and imports): $Y = C + I + G$
  - $S + T + C = Y = C + I + G$
  - $\implies I = S + (T - G)$ (with trade: $I = S + (T - G) + (IM - X)$)
- The goods market is only in equilibrium if investment equals savings (the sum of private and state savings).
- Is S or I the driver? Theory of effective demand (Keynes) versus supply-oriented theories (Say's law, (neo-)classical).

The IS Curve: Summary

- Implicit IS curve: $Y = C(Y - T) + I(Y, i) + G$ (assumption: $X = IM = 0$)
- The IS curve depends negatively on the interest rate:
  With increasing interest rates, investment demand and total production decreases.
  - Where does the interest rate come from?
    $\longrightarrow$ Financial Market, LM-curve, Lecture 3
- Movement along the IS curve:
  Interest rate change $\longrightarrow$ Endogenous adjustment of production volume
- Shifts in the IS curve:
  Factors that trigger a decrease (an increase) in the demand for goods at a given interest rate, shift the IS curve to the left (right).
  Example: increased government spending.

Summary and Outlook

- We analyzed the goods market of an economy.
- The IS curve summarizes the equilibrium on the goods market in the interest/income graph.
- We will introduce the LM curve in lecture 3.
- We will then combine IS and LM to discuss the IS-LM model in Lecture 4.
- Preparation for lecture 3: Slides and chapter 4 & 5.2 of the textbook

Discussion

- Open questions from the audience?
- The savings paradox — is saving bad?
  - Y decreases with increasing propensity to save (i.e. with smaller c_0, c_1)
  - The higher c_1 the higher the multiplier.

Application to the Current Situation

- Consumer sentiment in Switzerland, calculated by SECO, has continued to recover. What does this mean for the Swiss economy?